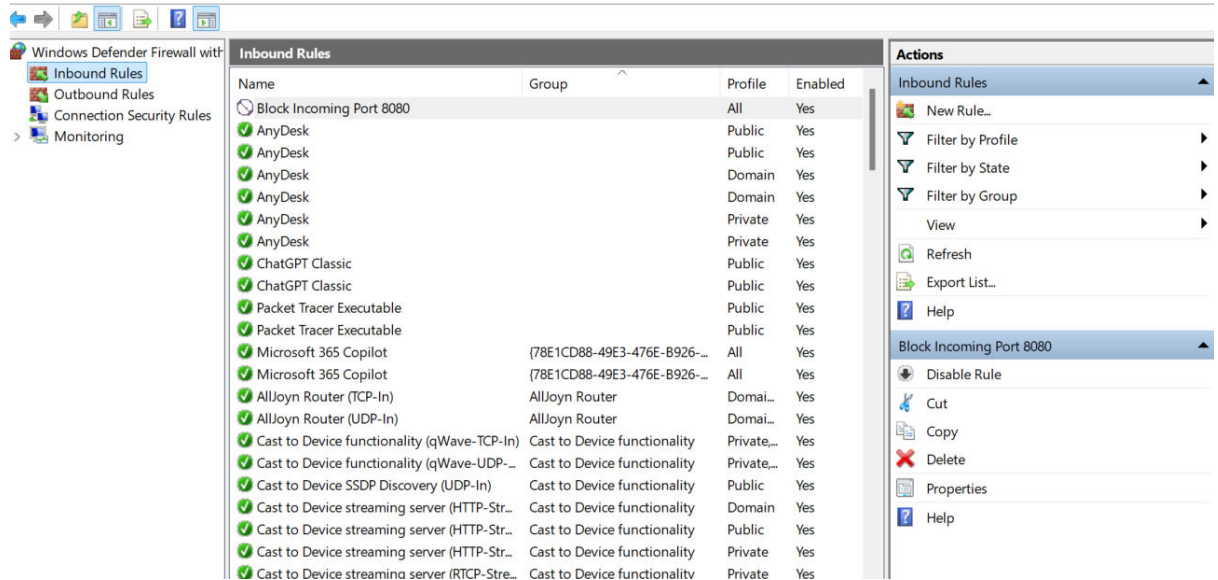


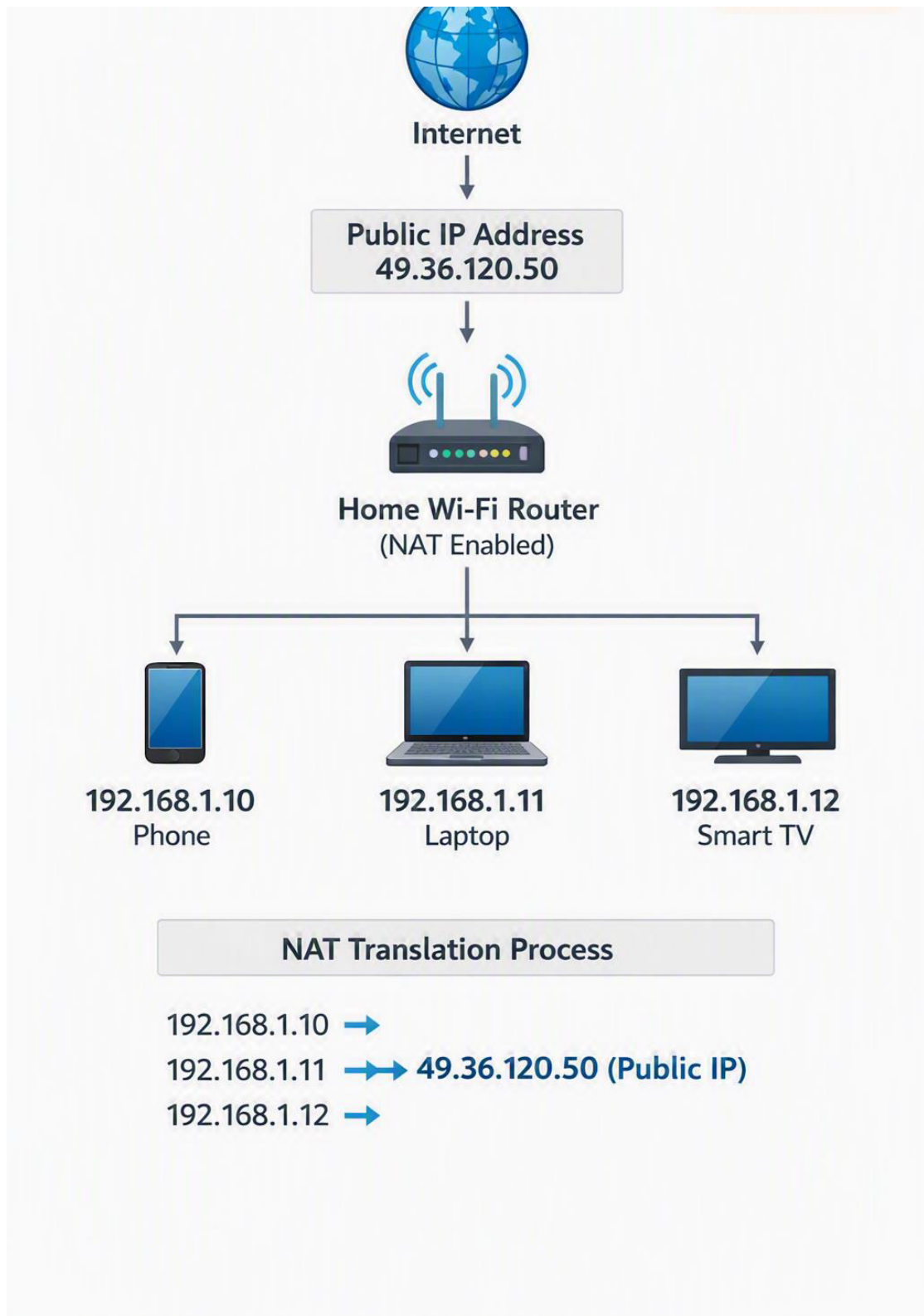
Tasks'

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.



3. **Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**



Steps to Configure Port Filtering to Block Outgoing Port 25

Step 1: Connect your laptop or phone to the home Wi-Fi network.

Step 2: Open a web browser and enter the router's default IP address, such as 192.168.1.1.

Step 3: Login to the router admin panel using the administrator username and password.

Step 4: Go to the Security Settings / Firewall / Access Control / Port Filtering section.

Step 5: Enable Port Filtering and create a new rule.

Step 6: Enter the following details:

- Rule Name: Block SMTP Port 25
- Protocol: TCP
- Direction: Outgoing
- Port Number: 25
- Action: Block/Deny

Step 7: Save the settings and restart the router if required.

4. **Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.**



Note: The answer depends on the rule set provided by your trainer. Without the rule set, it is not possible to determine whether these packets are allowed or blocked.

Sample packets:

1. HTTP: 192.168.1.5 → 172.217.166.78:80

2. SSH: 10.0.0.10 → 34.201.23.1:22

3. DNS: 192.168.1.7 → 8.8.8.8:53

Upload or share the packet filtering rule set, and I can tell you which packets are Allowed or Blocked

5. **Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.**



- Block Unauthorized Access: Firewalls block hackers from accessing servers through unused or dangerous ports.
- Allow Only Required Traffic: Packet filtering allows only trusted traffic (such as HTTPS on port 443) and blocks unwanted connections.

- **Protect Against Attacks:** Firewalls detect and block suspicious traffic such as DDoS attacks or repeated malicious login attempts, helping keep the service available for users